Normalized amplitude features: 12 equations

PCG-BP analysis version 2 | Feature calculation specification

For each complete adjacent S1-S2-S1 cycle, a, b, c, d, e denote S1 onset, S1 offset, S2 onset, S2 offset and next S1 onset, in zero-based recording samples. They must satisfy a < b < c < d < e. Sampling frequency is f_s. States use half-open sample intervals [a,b), [b,c), [c,d), [d,e).

Only complete cycles contained in a segment contribute. The next S1 onset may equal the exclusive segment end; its waveform belongs to the next segment. Detection is performed once over the whole recording.

Normalize the filtered waveform separately within each saved segment to zero mean and unit population SD. L is the number of segment samples. A_Q is the mean absolute normalized waveform amplitude over state Q, not an envelope peak or RMS. Exclude a cycle if any state amplitude is at most 1e-12. All six ratios use the same valid-cycle pool.

$$z[n] = \frac{x[n] - \mu_x}{\sigma_x}, \quad \mu_x = \frac{1}{L}\sum_n x[n], \quad \sigma_x = \sqrt{\frac{1}{L}\sum_n (x[n] - \mu_x)^2}$$

$$A_{Q,i} = \frac{1}{|I_{Q,i}|} \sum_{n \in I_{Q,i}} |z[n]|$$

Per-cycle quantity	Equation	Output columns
Amp_SysS1	$A_{Sys,i}/A_{S1,i}$	m_Amp_SysS1 sd_Amp_SysS1
Amp_DiaS2	$A_{Dia,i}/A_{S2,i}$	m_Amp_DiaS2 sd_Amp_DiaS2
Amp_S1S2	$A_{S1,i}/A_{S2,i}$	m_Amp_S1S2 sd_Amp_S1S2
Amp_S1Dia	$A_{S1,i}/A_{Dia,i}$	m_Amp_S1Dia sd_Amp_S1Dia
Amp_SysDia	$A_{Sys,i}/A_{Dia,i}$	m_Amp_SysDia sd_Amp_SysDia
Amp_S2Sys	$A_{S2,i}/A_{Sys,i}$	m_Amp_S2Sys sd_Amp_S2Sys

Aggregation within each segment

$$m_q = \frac{1}{K}\sum_{i=1}^{K} q_i$$

$$sd_q = \sqrt{\frac{1}{K-1}\sum_{i=1}^{K} (q_i - m_q)^2}$$

Ratios are calculated per cycle, then averaged. Sample SD uses K-1 (an explicit implementation convention). Mean is NaN for K=0; SD is NaN for K<2. Invalid or ambiguous cycles are excluded, never bridged or filled with zero.

Source: Tang et al. (2018), PCG Classification Using Multidomain Features and SVM Classifier, Tables 1-2 and Sections 2.3-2.5.3. DOI: 10.1155/2018/4205027. Feature definitions follow the paper; this project retains its 30-500 Hz Butterworth filter and timing-based detector rather than the paper's 10 Hz high-pass, spike removal and Springer HSMM.